Permutation Test

Statistical Computing, STA3005

April 15, 2026

Last chapter: Simulation and Bootstrap

Part I: *Motivation*

Permutation tests are based on **resampling**, but unlike bootstrap, the samples are drawn **without replacement**.

Suppose we have two groups of samples $X_1, X_2, \ldots, X_n \sim F$ and $Y_1, Y_2, \ldots, Y_m \sim G$

Want to do the test without specifying the parametric forms of F and G

$$H_0 : F = G \quad \text{vs} \quad H_1 : F \neq G,$$

Example

Compare the median of samples from two populations. Based on the observed data, we can compute some test statistics

- $T_{obs} = \frac{median(X_1, X_2, \ldots, X_n)}{median(Y_1, X_2, \ldots, Y_m)}$
- $T_{obs} = median(X_1, X_2, \ldots, X_n) - median(Y_1, X_2, \ldots, Y_m)$
- $T_{obs} = |median(X_1, X_2, \ldots, X_n) - median(Y_1, X_2, \ldots, Y_m)|$

How could we know the null distribution of T?

Permutation distribution

Let Z be the ordered set $\{X_1, \ldots, X_n, Y_1, \ldots, Y_m\}$

- $Z_i = X_i$ if $1 \leq i \leq n$
- $Z_i = Y_{i-n}$ if $n + 1 \leq i \leq n + m$.

Let $Z^* = (X^*, Y^*)$ represent a partition of the pooled sample $Z = X \cup Y$, where X^* has n elements and Y^* has $N - n = m$ elements.

- Z^* corresponds to a permutation π of Z such that $Z_i^* = Z_{\pi(i)}$
- $\binom{N}{n}$ different ways to partition the pooled sample Z into two subsets of size n and m.

If $T(X, Y) = T(Z)$ is a statistic, then the permutation distribution of T^* is equivalent to draw from

$$\left\{T_j^* = T(Z_j^*) = T(Z_{\pi_j}) \mid \pi_j \text{ is a permutation of } m+n \text{ samples}, j = 1, \ldots, \binom{N}{n}\right\}.$$

with the same probability.

Example

Test statistics for the permutated samples Z^*

- $T^* = \frac{median(Z_1^*, Z_2^*, \ldots, Z_n^*)}{median(Z_{n+1}^*, Z_{n+2}^*, \ldots, Z_{n+m}^*)}$
- $T^* = median(Z_1^*, Z_2^*, \ldots, Z_n^*) - median(Z_{n+1}^*, Z_{n+2}^*, \ldots, Z_{n+m}^*)$
- $T^* = |median(Z_1^*, Z_2^*, \ldots, Z_n^*) - median(Z_{n+1}^*, Z_{n+2}^*, \ldots, Z_{n+m}^*)|$

p-value based on permutation distribution

If $T(Z) = |median(Z_1, Z_2, \ldots, Z_n) - median(Z_{n+1}, Z_{n+2}, \ldots, Z_{n+m})|$, then the p-value of the observed statistic T_{obs} is

$$p = \Pr(T^* \geq T_{obs}) = (\binom{N}{n})^{-1} \sum_j I(T_j^* \geq T_{obs}).$$

However, in reality, it is impossible to compute all $\binom{N}{n}$ permutations.

Part II: *Approximate permutation test procedure*

Instead, an approximate permutation test is implemented by randomly drawing a large number of samples without replacement.

1. Compute the observed test statistic $T_{obs} = T(Z)$.
2. For each replicate, indexed $b = 1, \ldots, B$:
    a. Generate a random permutation π_b.
    b. Compute the statistic $T_b^* = T(Z_b^*)$.

3. If large values of T_{obs} support the alternative, compute the empirical p-value by

$$\hat{p} = \frac{\#\{T_b^* \geq T_{obs}\}}{B} = \frac{1}{B}\sum_{b=1}^{B} I(T_b^* \geq T_{obs}).$$

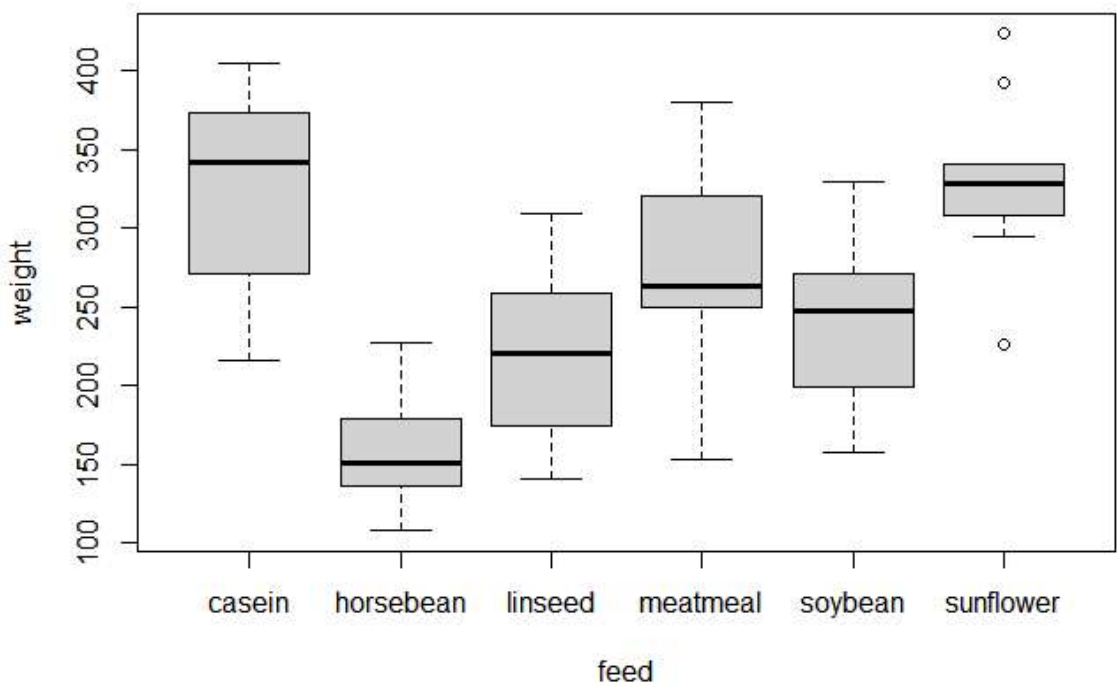
   For a lower-tail or two-sided test $\hat{p}$ is computed in a similar way.
4. Reject H_0 at significance level α if $\hat{p} \leq \alpha$.

It is suggested that 100 to 1,000 permutations should suffice.

Example

The `chickwts` data in `R` records the weights in grams for six groups of newly hatched chicks fed different supplements. There are six types of feed supplements. Let's display a graphical summary of the data

```
boxplot(weight ~ feed, chickwts)
```



The plot suggests that `soybean` and `linseed` groups may be similar. The distributions of weight for these two groups are compared below.

```
x <- sort(chickwts$weight[chickwts$feed == "soybean"])
y <- sort(chickwts$weight[chickwts$feed == "linseed"])
```

Let consider four test statistics:

- t statistics
- median ratio
- median difference
- absolute median difference

```
set.seed(1234)
R <- 1000 #number of replicates
z <- c(x, y) #pooled sample
K <- 1:length(z)
reps <- matrix(NA, R, 4) #storage for replicates

t0 <- c(t.test(x, y)$statistic,
        median(x)/median(y),
        median(x) - median(y),
        abs(median(x) - median(y)))

for (i in 1:R) {
  # generate indices k for the first sample
  k <- sample(K, size = 14, replace = FALSE)
  x1 <- z[k]
  y1 <- z[-k]
  t1 <- c(t.test(x1, y1)$statistic,
              median(x1)/median(y1),
              median(x1) - median(y1),
              abs(median(x1) - median(y1)))

  # record the test statistics for permuted samples
  reps[i,] <- t1
}

# p-val of t statistics
t0[1]
```

```
##        t
## 1.324556
```

```
p_t <- sum(abs(reps[,1]) >= abs(t0[1]))/R
p_t
```

```
## [1] 0.191
```

```
# p-val for median ratio
t0[2]
```

```
##
## 1.122172
```

```
p_rat <- sum(reps[,2] >= t0[2] | reps[,2] <= 1/t0[2])/R
p_rat
```

```
## [1] 0.281
```

```
# p-val for median difference
t0[3]
```

```
##
## 27
```

```
p_dif <- sum(reps[,3] >= t0[3] | reps[,3] <= -t0[3])/R
p_dif
```

```
## [1] 0.283
```

```
# p-val for absolute median difference
t0[4]
```

```
##
## 27
```

```
p_absdif <- sum(reps[,4] >= t0[4])/R
p_absdif
```

```
## [1] 0.283
```

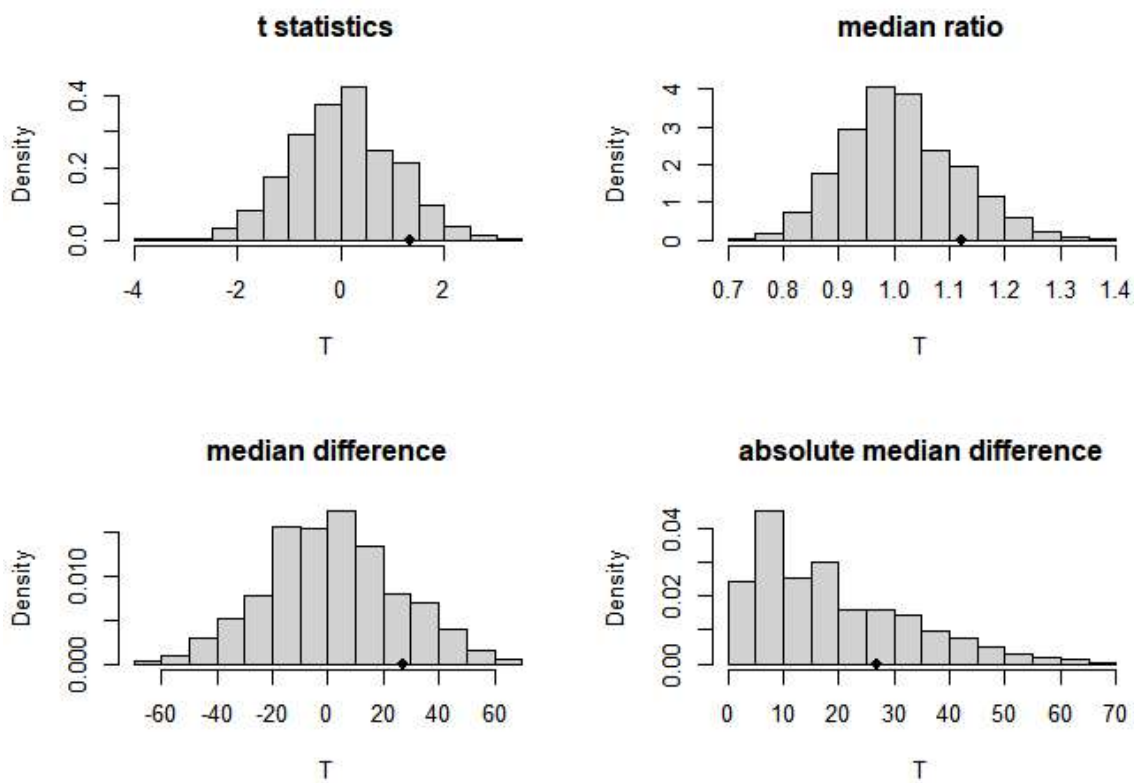
In all of the four tests, we cannot reject the null hypothesis. Now, let's visualize the approximated permutation distribution

```
par(mfrow = c(2,2))
hist(reps[,1], main = "t statistics", freq = FALSE, xlab = "T", breaks = "scott'
points(t0[1], 0, cex = 1, pch = 16) #observed T

hist(reps[,2], main = "median ratio", freq = FALSE, xlab = "T", breaks = "scott'
points(t0[2], 0, cex = 1, pch = 16) #observed T

hist(reps[,3], main = "median difference", freq = FALSE, xlab = "T", breaks = "s
points(t0[3], 0, cex = 1, pch = 16) #observed T

hist(reps[,4], main = "absolute median difference", freq = FALSE, xlab = "T", br
points(t0[4], 0, cex = 1, pch = 16) #observed T
```



Part III: *Tests for Equal Distributions*

Instead of comparing sample mean or sample median, we can choose a test statistic that measures the difference between two distributions. We still consider the hypothesis

$$H_0 : F = G \text{ vs } H_1 : F \neq G.$$

Two-sample Kolmogorov-Smirnov (K-S) statistic

The K-S statistic D is the maximum absolute difference between the ECDF's of the two samples, defined by

$$D = \sup_{1 \le i \le N} |F_n(z_i) - G_m(z_i)|,$$

where

- F_n is the ECDF of the first sample $x_1, \ldots, x_n$
- G_m is the ECDF of the second sample $y_1, \ldots, y_m$.
- Here, $0 \le D \le 1$ and large values of D support the alternative $F \neq G$.

Example

Going back to the chick weight example, F is the distribution of weight of chicks fed `soybean` supplements and G is the distribution of weight of chicks fed `linseed` supplements.

```
D0 <- ks.test(x, y, exact = FALSE)$statistic
```

```
## Warning in ks.test.default(x, y, exact = FALSE): p-value will be approximate
## the presence of ties
```

```
D0
```

```
##        D
## 0.297619
```

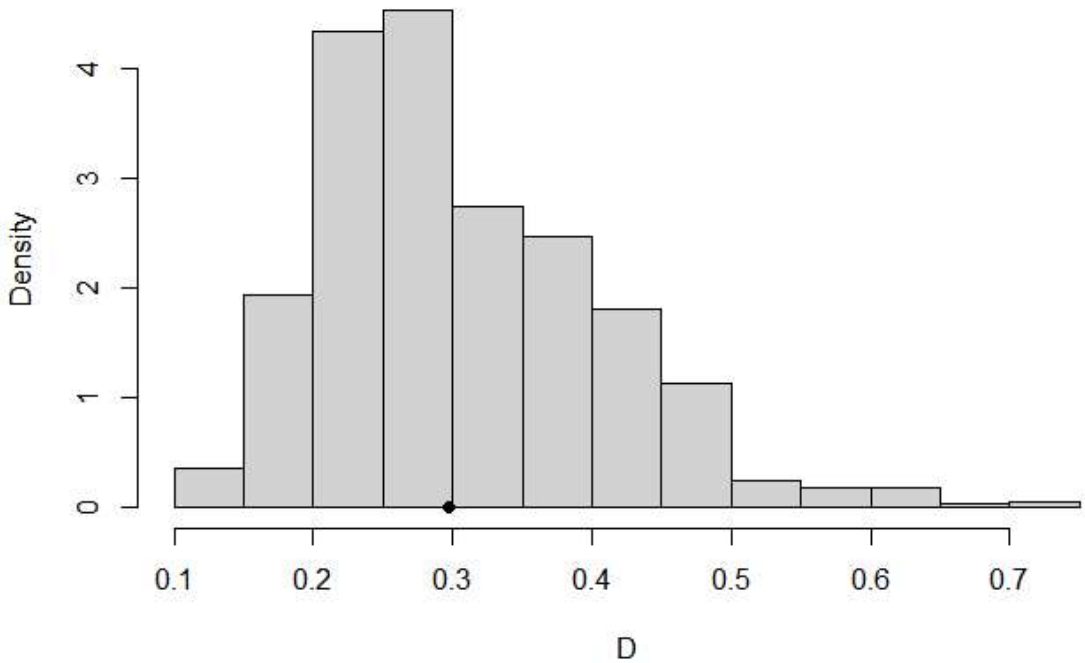
The observed value of $D = D(Z)$ can be computed using `ks.test`. Note that the K-S test `ks.test` generates a warning each time it tries to compute a p-value, because there are ties in the data. We are not using the p-value, so it is safe to ignore these warnings.

To determine whether this value of D is strong evidence for the alternative, we compare D with the replicates $D^* = D(X^*, Y^*)$.

```
D <- rep(NA, R) #storage for replicates
options(warn = -1)
```

```
for (i in 1:R) {
  # generate indices k for the first sample
  k <- sample(K, size = length(x), replace = FALSE)
  x1 <- z[k]
  y1 <- z[-k] #complement of x1
  D[i] <- ks.test(x1, y1, exact = FALSE)$statistic
}
p <- mean(D >= D0)

# Visualize the approximated permutation distribution
hist(D, main = "", freq = FALSE, xlab = "D",
breaks = "scott")
points(D0, 0, cex = 1, pch = 16) #observed D
```



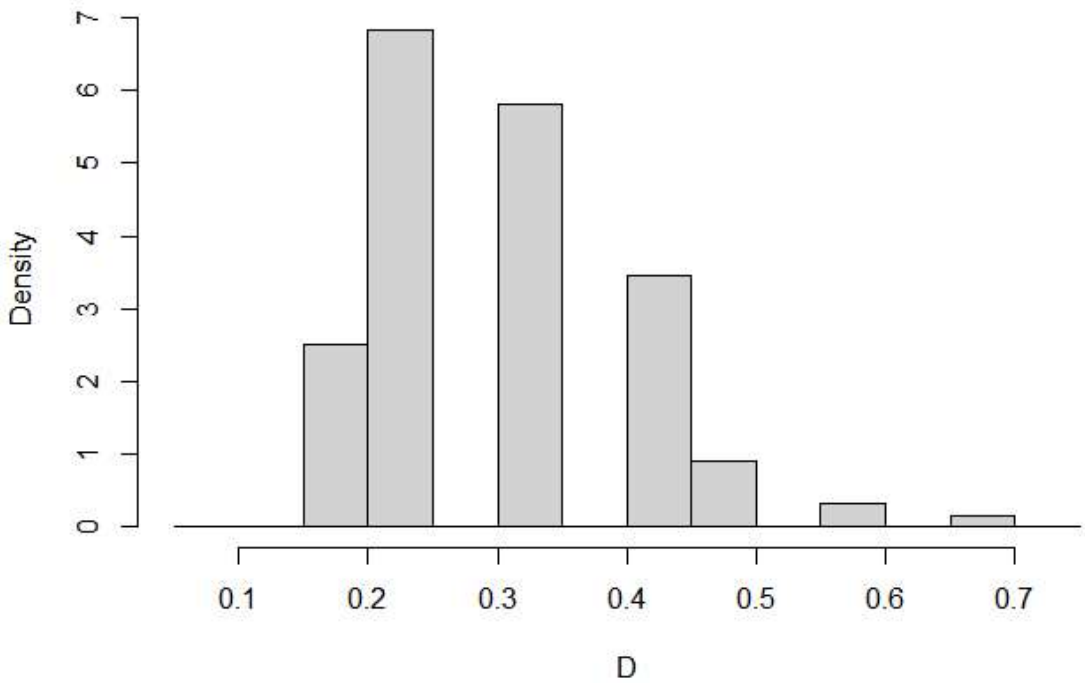
Do a comparison of chick weights fed `sunflower` and `linseed` supplements

```
x <- sort(chickwts$weight[chickwts$feed == "sunflower"])
y <- sort(chickwts$weight[chickwts$feed == "linseed"])

z <- c(x, y) #pooled sample
K <- 1:length(z)
D <- numeric(R) #storage for replicates
options(warn = -1)
D0 <- ks.test(x, y, exact = FALSE)$statistic
for (i in 1:R) {
```

```
#generate indices k for the first sample
k <- sample(K, size = length(x), replace = FALSE)
x1 <- z[k]
y1 <- z[-k] #complement of x1
D[i] <- ks.test(x1, y1, exact = FALSE)$statistic
}
p <- mean(D >= D0)

# Visualize the approximated permutation distribution
hist(D, main = "", freq = FALSE, xlab = "D", breaks = "scott")
points(D0, 0, cex = 1, pch = 16) #observed D
```



Permutation test and bootstrap

The permutation test and the bootstrap are both nonparametric resampling methods for statistical inference.

Feature	Permutation Test	Bootstrap
Target	Hypothesis testing (e.g., $F = G$)	Bias, Confidence intervals (CI), Standard error (SE) estimation
Resampling	Sample without replacement	Sample with replacement

Feature	Permutation Test	Bootstrap
Intuition	Approximate the null distribution by resampling	Approximate the population distibution by empirical distribution
Outcome	p-value for the hypothesis	Bias, SE, CI for parameters of interest

Summary

- Goal: Conduct hypothesis testing between two groups of samples without requiring distribution assumption
- Define a test statistics and then approximate the null distribution by permuting or shuffling all of the samples
- The empirical p-value based on B random permutations

$$\hat{p} = \frac{1}{B}\sum_{b=1}^{B} I(T_b^* \geq T_{obs}).$$

- Two-sample Kolmogorov-Smirnov statistic that measures the difference between two distributions